PRESINT'S OFFICE.

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT.

ARUSHA REGION.

FORM FOUR MOCK EXAMINATION.

BIOLOGY 01.

MARKING SCHEME.

SECTION A (16 Marks).

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	C	C	D	B	B	B	C	B	D

(@01 = 10marks).

2.

i	ii	iii	iv	v	vi
D	F	E	G	B	C

(@01 = 06marks).

3. (a) Mitosis. (1 mark).

Mitosis is a Cellular division in Somatic Cells resulting into two diploid daughter Cells. (2 marks).

(b) Significance of Mitosis.

(i) Facilitates growth by increasing the number of Cells.

(ii) Facilitates Cell repair and replacement.

3. (b) (iii) Regeneration of the lost parts in So me organisms.

(iv) Maintains genetic stability.

(v) It is a means of reproduction in uni Cellular organisms and those organism that reproduce asexually.

(Any four (4) points @ 1.5 marks = 06 marks)

Total = 09 marks

4. (a) – Enhancing nutrient availability (In Crease Soil fertility).

– Promoting decomposition (They are decomposers).

– Increase Soil aeration by loseni ng the Soil.

– Improve water holding Capacity.

(1 @ = 3 marks).

(Any three (3) points).

(b) They possess both male and female reproductive organs in an individual worm.

2 marks.

(c) – They posses exoskeleton.

– They posses appendages.

– They have high reproductive potenti al.

– They have segmented body.

– Adaptability to different environme nt.

(Any four (4) points. 1 @ = 4 marks)

Total = 09 marks

5. (a) Irresponsible Sexual behaviours characterized by a lack of Consideration for potential Consequences and harm while Responsible Sexual behaviours prioritize health, well-being and respect for others.

3 marks.

(b) Four ways which eradicate irresponsible sexual behaviours.

- Focus on education.
- Promote health relationships.
- Support system.
- Strengthening Consent.
- Have boundaries.
- Be principled.

(Any four (4) points. $0 \frac{1}{2}$ @ = 6 marks)

Total = 09 marks

6. (a) (i) Glucose. (01 mark).

(ii) Insulin. (01 mark).

(iii) The role of insulin is to convert glucose to glycogen and stimulates the uptake of glucose to the body cells for respiration. ($02 \frac{1}{2}$ marks).

(iv) Diabetes Mellitus. (01 mark).

(v) Symptoms of Diabetes Mellitus.

→ Hyperglycaemia (High level of glucose in the blood.

→ Glycosuria (Excretion of large amount of glucose in urine).

6. (a)(v) → Diuresis - The production of large amount of glucose in urine.

→ Constant thirst.

→ The patient becomes thin and weak.

Any one (1) point = (01 mark).

~~Total = 09 marks~~.

(b) Changes/stages needed for Seed germination.

(i) Seed absorbs water.
(ii) Digestion of the stored food.
(iii) Oxidation of food to produce energy.
(iv) Bursting of seed coat/testa.
(v) Plumule and radical emerge.

(@ $00\frac{1}{2}$ = $02\frac{1}{2}$ marks).

Total = 09 marks

7. (a)

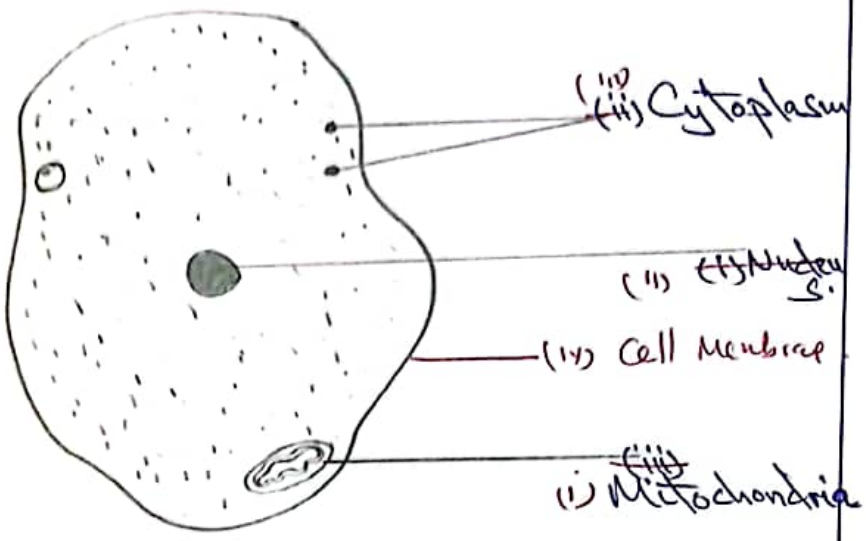
Prokaryotic Cells	Eukaryotic Cells.
(i) Lack true nucleus.	- Have true nucleus.
(ii) Have no membrane bound organelles.	- Have membrane bound organelles.
(iii) Small in size.	- Large in size.
(iv) Have Circular DNA	- Have linear DNA.

Any three (3) points, @1 = 3 marks.

(b) - chloroplast.
- Cell wall.
- Large and permanent vacuole.

Any two (2) points, @1 = 2 marks

07. (c) Diagram of animal Cell.



C — 00 ½ mark.

D — 02 mark.

L — 3 pts @ 00½ = 01½ marks

Total = 09 marks.

8. (a)(i) It is healthier to breath through the nose than through the mouth: Because nose filters out dusts, humidifies and warms the air before it enters the lungs which helps prevent infections, improves oxygen absorption and maintains a healthy respiratory system. (01½ marks).

(ii) During digestion the food is acidic when in the stomach due to production of gastric juice which mainly contains hydrochloric acid, ideal for the optimum activity of the stomach enzymes. (01½ marks)

8 (b) Significance of Transpiration to plants

(i) Helps to maintain transpiration pull which is important for maintaining a constant stream of water between the roots and the leaves.

(ii) It enables the loss of excess water from the plant.

(iii) It enables absorption and distribution of water and mineral salts.

(iv) It helps to cool the plant.

(Any 3 pts. @ 01 = 03 marks).

(c) Role of skeleton.

(i) Movement / Locomotion. i.e Facilitate movement.

(ii) Protection of delicate organs / internal organs.

(iii) Gives the body shape.

(iv) Storage of minerals.

(v) Manufacture of blood cells.

(vi) Supports the body.

(Any 3 pts. @ 01 = 03 marks)

9. Introduction – Any related introduction for example
Bacteria are single celled organisms which are found almost everywhere on the earths surface.
(01½ marks)

Main body – Importance of bacteria.
(i) Some are used in making antibiotics.
(ii) They are useful in nitrogen fixation.
(iii) Some produce lactic acid during fermentation.
(iv) Some bacteria are autotrophs.
(v) Some are used in making biogas.
(vi) Some used in production of flavour in sour milk, cheese, Yoghurt.
(vii) They help in decomposition of organic matters to clean environment and ensure Circular nutrients in an ecosystem.
(viii) Used in genetic engineering to produce hormones like insulin and growth hormones.
(Any six(6) points @½=3 marks)

Harmfulness – Some cause infections and diseases.
– Some destroy the crops.
– Some cause food spoilage
– Denitrifying bacteria reduce soil fertility
(Any six(6) points @½=3 marks)

Any related Conclusion – 1½ marks.

10. (a) Green seeds was dominant.
- This is because all F1 offsprings has green seeds. 02 marks.

(b) To show how F1 individuals were obtained. (green seeds).

Let G - Represent gene for green seeds. (00½ mrk)
g - Represent gene for yellow seeds. (00½ mrk)

Then; 1st Cross ÷ Cross between Green seeds and Yellow seeds.

Parental phenotype: (01 mrk) Green seeds x yellow seeds

Parental genotype: (01 mrk) GG x gg.

Meiosis: (01 mrk)

Gametes: (00½ mrk) G G g g

Fertilization: (00½ mrk)

F1 genotype: (00½ mrk) Gg Gg Gg Gg

F1 phenotype: (00½ mrk) All were Green seeds.

(c) 2nd Cross ÷ Selfing F1.

10. (c).

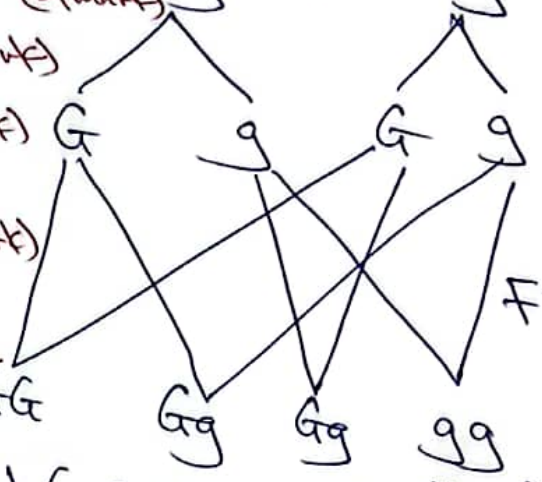
parental phenotype: Green seed × Green seed. (01 mark)

parental genotype (01 mark) Gg × Gg.

meiosis: (01 mark)

Gametes: (01 mark) G g G g

fertilization: (01 mark)



F2 genotype: 01/2 mark GG Gg Gg gg

F2 phenotype: 3:1 (3 green : 1 yellow) 01/2 mark

(d) Homozygous green.

$\frac{1}{4} \times 312 = 78$. (1 mark).

Total = 15 marks.

11. Introduction — 02 marks.

Main body — Giving balanced diet.
— Taking to Centres that provide HIV and AIDS information and medical care.
— Provide ARV drugs.
— Allow the rest when feel unwell.
— Allow to work and exercise if they can.
— Avoid discriminating/stigmatising.
— Counselling.
(Any six (6) points. @ 02 = 12 marks)

Conclusion — 1 mark.

Total = 15 marks.

— 09 —

THE PRESIDENT'S OFFICE.
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT.
ARUSHA REGION.
FORM IV MOCK EXAMINATION.
033/2A. BIOLOGY-2A. - MAY-2025
(ACTUAL PRACTICAL).
MARKING SCHEME.

(8)

1. (i) To test the presence of Starch in a given food Substance. (03 marks).
(ii) The food Substance (Specimen M) change to blue black/dark blue. (03 marks).
(iii) Starch was present. (02 marks).
(iv) Cassava/Sweet potato/Cocoyam/Banana
Any 2 Sources.
(@ 01 mark = 02 marks)
(v) Source of energy in the body (02 marks)
(vi) Mouth, Duodenum, Ileum
(02 marks @ = 06 marks).
(vii) Liver and Muscles. (02 marks @ = 04 marks)
(viii) Excess glucose is Converted to glycogen.
(03 marks)

Total = 25 marks.

2. (a) L — Grasshopper. — 0/mark.
B — Mushroom. — 0/mark.
K — Lizard. — 0/mark.

(b)

Specimen	kingdom	phylum.	class.
L	Animalia 0/mark	Arthropoda 0/mark	Insecta 0/mark
B	Fungi 0/mark	Basidiomycota 0/mark	Agaricomycetes 0/mark
K	Animalia 0/mark	Chordata 0/mark	Reptilia. 0/mark

(c) Specimen L:
Trophic level: Primary Consumer. 0/mark
Reason: They are herbivores. i.e feed directly on plants. 0/mark

Specimen B.
Trophic level: Decomposers. 0/mark
Reason: They breakdown dead organic matter and recycle nutrients back to the ecosystem. 0/mark

Specimen K.
Trophic level: Secondary Consumer. 0/mark
Reason: They are Carnivores, they feed on insects. 0/mark

2. (d) Economic importance of specimen B.

- They are saprophytes hence increase soil fertility.
- Can be used for production of starch.
- used in scientific investigation.
- They can be used as food for humans.

(Any one point). (02marks).

(e) Adaptive features of specimen B.

- Has stipe that support the cap so that the dispersal of spores occur easily.
- They have hyphae help absorption of water and mineral nutrients.
- Some species are poisonous, this help them from being eaten by predators.
- They have cap for protection of basidio spores.
- Have abilities to secrete enzymes of extracellular digestion.

Any five points. @1 = 05marks.

Total = 25marks.

THE PRESIDENT'S OFFICE.
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT.
ARUSHA REGION.
FORM FOUR MOCK EXAMINATION
033/2B. BIOLOGY-2B. MAY-2025.
(ACTUAL PRACTICAL).

MARKING SCHEME.

(a) FOOD TEST REPORT.

1.

FOOD TESTED	PROCEDURE	OBSERVATION	INFERENCE
Starch 00 1/2	To 2Cm³ of Solution K in a test tube, three drops of Iodine solution were added then shaken. 01	The mixture turned into brown colour of Iodine Solution. 01	Starch was absent. 00 1/2
Protein 00 1/2	To 2Cm³ of solution K in a test tube, 2Cm³ of Sodium hydroxide solution was added followed by three drops of 1% Copper II Sulphate solution, then shaken. 01	The blue colour of Copper II Sulphate solution was retained. 01	Protein was absent 00 1/2
Lipids. (Fat and oil) 00 1/2	To 2Cm³ of Solution K in a test tube, three drops of Sudan III Solution were added then shaken vigorously and left to settle for five minutes. 01	The Solution form a red homogenous mixture with Sudan III Solution. 01	Lipid was absent 00 1/2
Reducing Sugar. 00 1/2	To 2Cm³ of Solution K in a test tube, 2 Cm³ of Benedict's Solution was added then boiled. 01	The series of Colour changed from blue, green, yellow, orange finally to brick red observed. 01	Reducing sugar was present 00 1/2

FOOD TESTED	PROCEDURE	OBSERVATION	INFERENCE
Non-reducing Sugar. 00 1/2	To 2cm³ of Solution K in a test tube, 1cm³ of distilled Hydrochloric acid was added then shaken and boiled for two minutes then Cooled, followed by addition of 1cm³ of Sodium hydroxide and 2cm³ Equal amount of Benedict's Solution then boiled. 01	The Series of Colour changed from blue green, yellow, orange finally to brick red 01	Non-reducing Sugar was present 00 1/2

(b) — Reducing Sugar.
— Non-reducing Sugar. @ 01 = 02 marks.

(c) — They provide energy (Sources of energy to the body). 01 mark.

(d) — Proteins.
— Vitamins.
— Lipids (fat and oil).
— Starch.
— minerals.
— Roughage.
— water. Any six pts. @ 00 1/2 = 03 marks

(e) — part in alimentary Canal.
• Ileum. — 01 marks.
Enzymes used
• Maltase.
• Lactase.
• Sucrase. 01 (@) = 03 marks.

2. (a) (i) Q- millet.

S- Earthworm.

@ 01 mark = 02 marks

(ii) R- classification.

kingdom - Animalia.
Phylum - Arthropoda.
class - Arachnida.

@ 01 mark = 03 marks.

Q - classification.

kingdom - Plantae.
Division - Angiospermatophyta.
class - Monocotyledonae.

@ 01 mark = 03 marks

(b) (i) It ensures the flow of food in an ecosystem, hence flow of energy from one level to another. 03 marks.

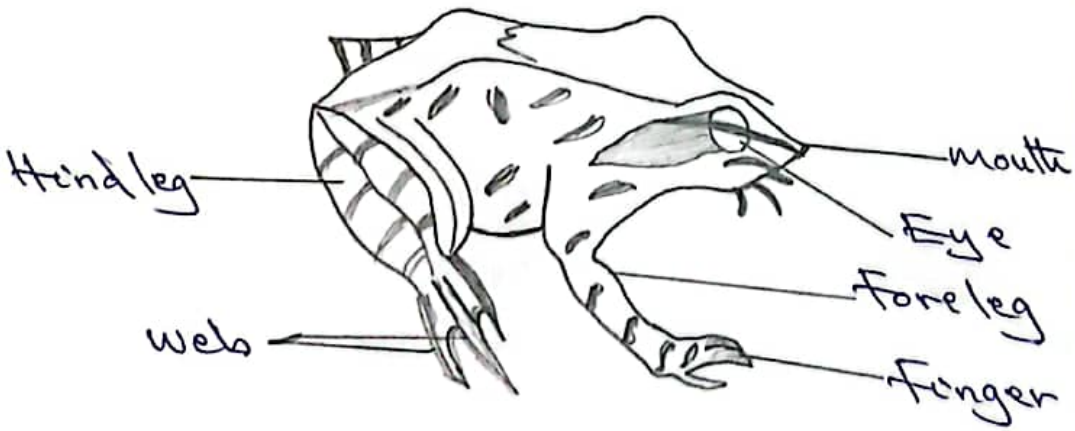


Diagram = 04 marks
Heading = 01 mark.
Any 5 labels = 05 marks.
total = 10 marks.

A diagram of specimen Q.

(c) It is important for specimens to support growth of plants due to the following ways;

- Improve soil aeration through burrowing in the soil.
- Increase soil moisture content through burrowing.
- Improve water drainage through burrowing.
- Improve soil fertility when they die and decompose.
- Left residue of their waste materials, add nitrogen and other humus in the soil.

Any 4 pts. @ 01 mark = 04 marks

Total = 25 marks.

THE PRESIDENT OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERMENT (8)

ARUSHA REGION

FORM IV-MOCK EXAMINATION

BIDLOGY - 2 C

ACTUAL PRACTICAL

033/2C MARKING SCHEME MAY 2025

1. (a) Diagram of External structure of human eye labelled with - Iris
- Pupil
- Eyelashes
- Eyelids
- Scleatic layer

Diagram = 02 marks
Heading = 01 mark
5 Labels = 0.5@ = 2.5 marks

Iris: Control the light entering the eye. 01/2 mark
Eyelids: Protect the eye from entering foreign bodies 01/2 mark
Scleatic layer: Protect the inner parts of the eye ball from mechanical injury 01/2 mark
Pupil: Allow the light to enter the eye. 01/2 mark
Eyelashes: Protect the eye from small particles such as dust 01/2 mark.

(b) Located in eye socket - 01 marks.

(c) For Vision - 02 marks.

(d) Adaptation

- Has eyelids protecting the eye from small particals and foreign object
- Eyelashes help to protect eye from small particles like dust
- Has sclerotic layer for protecting the inner part of Eyeball
- Has pupil regulating light to enter the eye

Any 4 pts. 01@ = 04 marks).

(e) (i) When the eye was covered with palm the pupil size of the eye wordened (dilated)

(ii) When the palm removed the pupil size of the eye decrease (constrict)

Any 5 points. 01@ = 05 marks.

Total = 25 marks.

2 ⓐ R - Hibiscus flower
S - Irish potato
T - Centipede
U - Millipede

0.5 ⓐ mark = 02 marks

ⓑ Classification

SPECIMEN	KINGDOM	PHYLUM/DIVISION	CLASS
R (Hibiscus	Plantae	Angiosperm	Dicotyledon
T (Centipede)	Animalia	Arthropoda	Chilopoda
U (Millipede)	Animalia	Arthropoda	Diplopoda

ⓐ 0.5 mark = 04 marks

ⓒ Structure of Specimen T and U

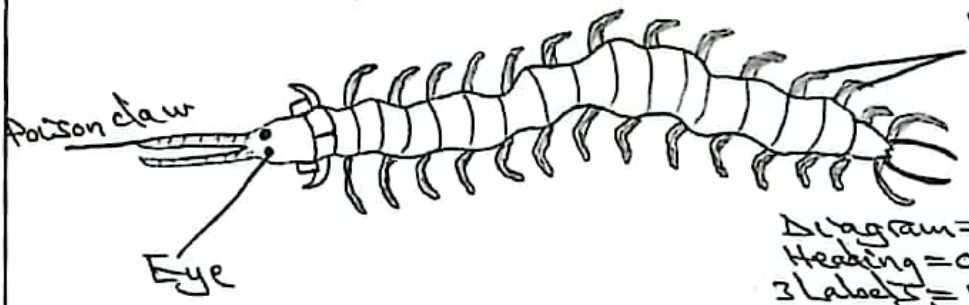


Diagram = 02 marks
Heading = 0.5 marks
3 Labels = 1.5 marks
Total = 04 marks

A well labelled diagram of specimen T.

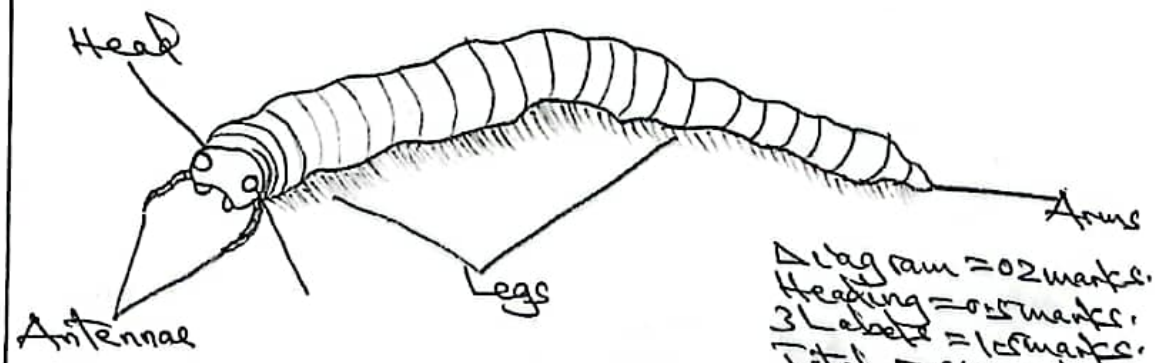


Diagram = 02 marks.
Heading = 0.5 marks.
3 Labels = 1.5 marks.
Total = 04 marks.

A well labelled diagram of specimen U.

d) Structure of Specimen R (Hibiscus flow ie longitudinal section

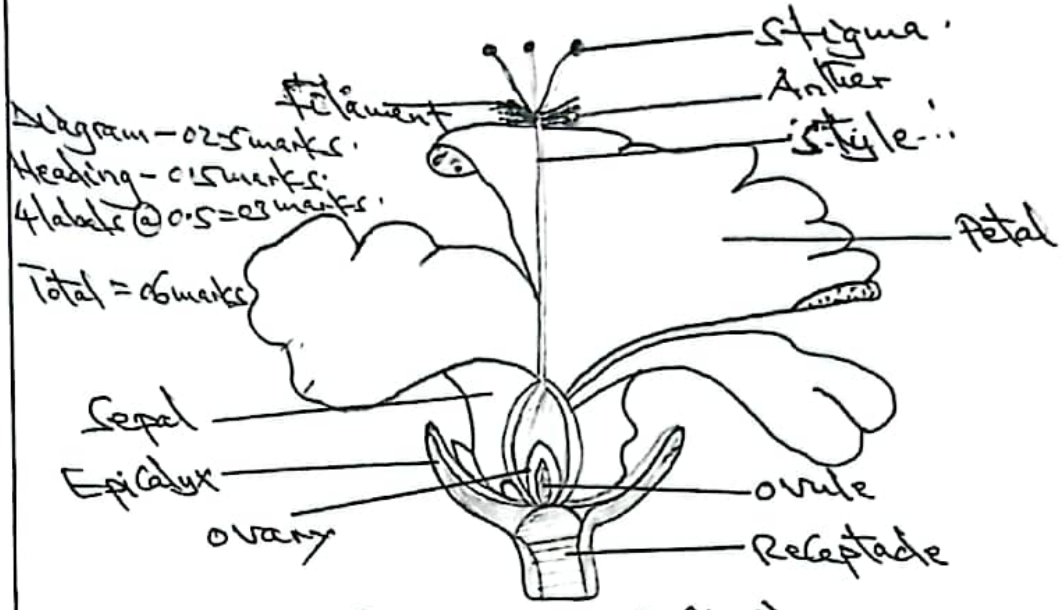


Diagram - 0.5 marks.
Heading - 0.5 marks.
4 labels @ 0.5 = 2 marks.
Total = 3 marks

A diagram of specimen R (L.S).

e) General characteristics of arthropods
- They have exoskeleton
- They have open circulation system
- They have jointed appendages
- They have segmented bodies
- Reproduction is mainly sexually
- The sexes are separated

Any 4 points @ 1 Mark

	f) Specimen T (Centipede)	U (Millipede)
I	Has one pair of legs per segment	Has two pair of legs per segment
II	Has flat body	Has cylindrical body
III	Has poisonous claws	Has no poisonous claw
	Any 2 points @	0 0½ 2